Qingyu Liu

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Education

Johns Hopkins University, Baltimore, MD, USA 2025.08 - present
Master of Science in Engineering (MSE) in Electrical and Computer Engineering

Shanghai Normal University, Shanghai, China 2020.10 - 2024.06
B.S. in Electronic Information Engineering

- **GPA**: **3.95**/4.00
- **TOEFL iBT**: **104**
- **Honors**: Second-Class Scholarship, Outstanding Graduate

Research Experiences

Multilingual TTS based on Flow Matching 2025.10 - present

- **Contributions**: Developing an open-source multilingual text-to-speech foundation model based on F5-TTS. Applying automatic speech recognition (ASR) to multilingual audio datasets without transcripts to generate transcriptions for large-scale training. Exploring scalable training strategies to handle multiple languages efficiently while maintaining high-quality speech synthesis across different linguistic characteristics.

Cross-Lingual F5-TTS 2025.03 - 2025.09

- **Contributions**: Developed a novel framework based on F5-TTS that enables cross-lingual zero-shot voice cloning without requiring audio prompt transcripts. Implemented MMS forced alignment to preprocess training data for identifying word boundaries, enabling direct synthesis from audio prompts while excluding transcripts during training. Designed and trained speaking rate predictors to derive duration from speaker pace. Achieved comparable performance to F5-TTS on LibriSpeech-PC test-clean and Seed-TTS-eval while successfully extending capabilities to cross-lingual voice cloning.
- **Outcomes**: **One paper** has been submitted to the International Conference on Acoustics, Speech and Signal Processing 2026 (ICASSP2026).

ICSD: An Open-source Dataset for Infant Cry and Snoring Detection 2023.11 - 2024.07

- **Contributions**: Constructed and open-sourced the **Infant Cry and Snoring Detection (ICSD) dataset** on GitHub for public research use. The ICSD dataset comprises three subsets: synthetic data created with Scaper, weakly labeled clips with event-level annotations, and strongly labeled actual data annotated manually. Proposed the Competitive CRNN+BEATs method, significantly improving system performance on the ICSD dataset.
- **Outcomes**: **One paper** has been submitted to the Journal of the Audio Engineering Society.

A Comparative Study of Deep learning Methods for Infant Cry Detection 2023.02 - 2023.10

- **Contributions**: Proposed two novel methods to improve baseline systems, including an ensemble learning method and a spectrogram-based preprocessing method applied in the CRNN and ASTs. These methods significantly enhanced system performance, achieving a 5.06% accuracy improvement on a noisy background test set compared to baseline models.
- **Outcomes**: **One paper** has been published in NCMMSC2023, poster presentation.

Paper List

- **Qingyu Liu**, Yushen Chen, Zhikang Niu, Chunhui Wang, Yunting Yang, Bowen Zhang, Jian Zhao, Pengcheng Zhu, Kai Yu, Xie Chen. “Cross-Lingual F5-TTS: Towards Language-Agnostic Voice Cloning and Speech Synthesis.” arXiv preprint.
- **Qingyu Liu**, Longfei Song, Dongxing Xu, Yanhua Long*. “ICSD: An Open-source Dataset for Infant Cry and Snoring Detection.” arXiv preprint.
- **Qingyu Liu**, Li Li, Yifan Zhou, Dongxing Xu, Yanhua Long*. “A Comparative Study of Deep Learning Methods for Infant Cry Detection.” in NCMMSC. 2023.

Awards

14th Challenge Cup, **Silver Award**	2024.07
China International College Students’ Innovation Competition (2024), **Silver Award**	2024.06
9th ”Think Youth” Digital Creation, Innovation & Entrepreneurship Competition, **First Prize**	2024.06
National Undergrad Training Program for Innovation and Entrepreneurship, **Project Leader**	2024.03

Internship

Shanghai Jiao Tong University X-Lance Lab | Research Assistant	2024.08 - Present
Unisound AI Technology Co., Ltd. | Research Intern	2023.11 - 2024.05